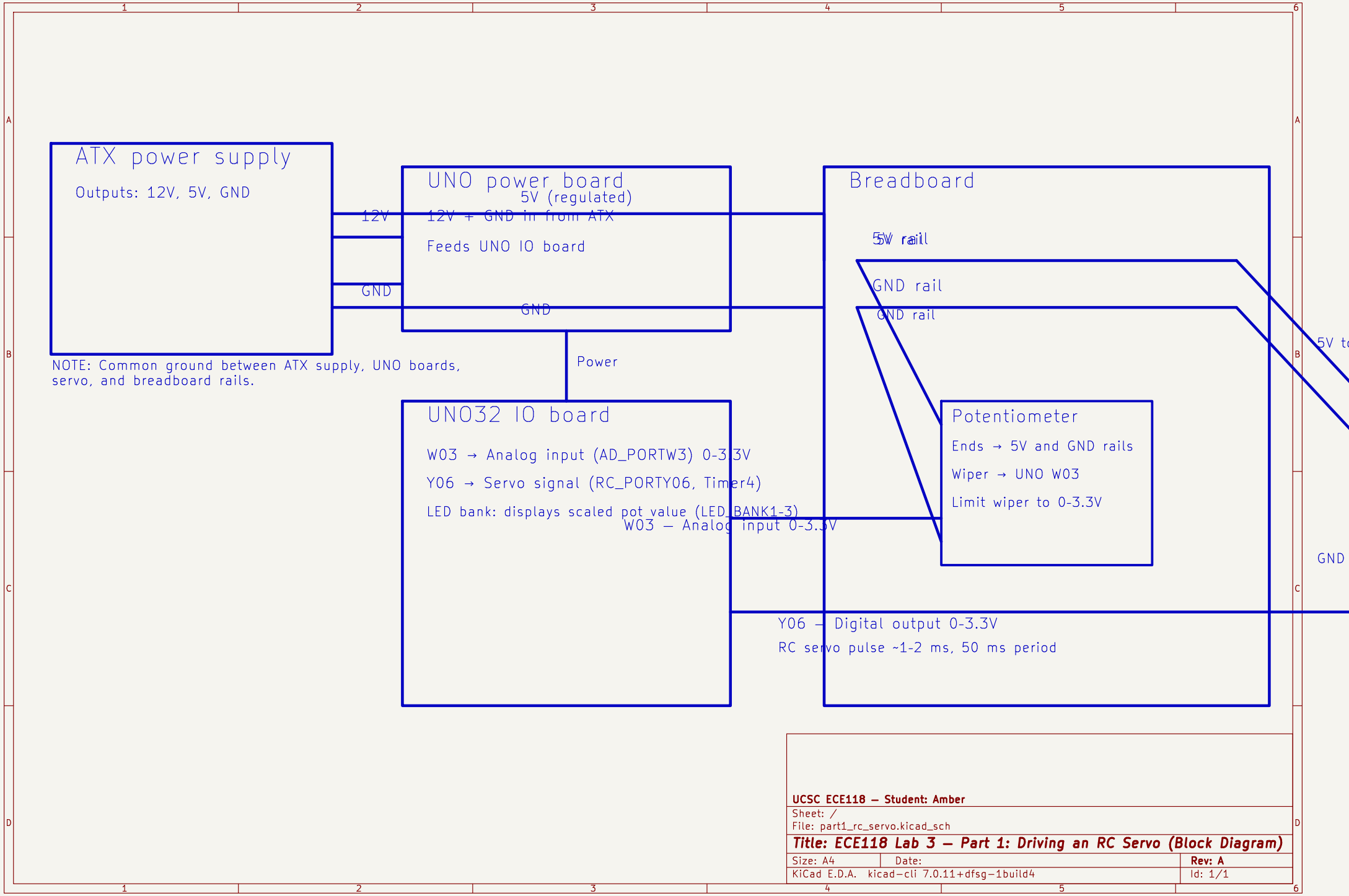
ATX power supply
Outputs: 12V, 5V, GND
12V
GND
UNO power board
5V (regulated)
12V + GND in from ATX
Feeds UNO IO board
GND
NOTE: Common ground between ATX supply, UNO boards, servo, and breadboard rails.
Power
UNO32 IO board
W03 → Analog input (AD_PORTW3) 0-3.3V
Y06 → Servo signal (RC_PORTY06, Timer4)
LED bank: displays scaled pot value (LED_BANK1-3)
Breadboard
5V rail
GND rail
GND rail
Potentiometer
Ends → 5V and GND rails
Wiper → UNO W03
Limit wiper to 0-3.3V
W03 — Analog input 0-3.3V
Y06 — Digital output 0-3.3V
RC servo pulse ~1-2 ms, 50 ms period
UCSC ECE118 — Student: Amber
Sheet: /
File: part1_rc_servo.kicad_sch
Title: ECE118 Lab 3 — Part 1: Driving an RC Servo (Block Diagram)
Size: A4
Date:
Rev: A
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